

# Ayush Kumar

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## SUMMARY

[portfolio](#) [twitter](#) [github](#)

Backend & GenAI Engineer focused on building scalable software systems, intelligent business platforms, and practical AI products. 13× Hackathon Winner recognized for delivering high-impact solutions through strong system design, full-stack engineering, and applied GenAI. Passionate about turning ambitious ideas into reliable, production-ready systems that solve real-world problems at scale.

## EXPERIENCE

### Zyoris Technology — AI Lead & Backend Engineer

#### AI-Powered CRM & Autonomous Business Operations Platform

Jul 2026 – Present

- Leading the design and development of AI-powered CRM workflows using Groq, OpenAI, Ollama, FastAPI, PostgreSQL, pgvector, BullMQ, and Next.js, enabling intelligent automation across sales, support, analytics, and operations.
- Architected RAG pipelines, vector memory systems, AI audit logging, PII redaction, autonomous agent workflows, structured data extraction services, and role-based AI copilots, supporting scalable enterprise-grade AI operations.

### Tech Trend Go — Software Development Engineer Intern

#### Full-Stack Development & Product Engineering

Jun 2026 – Jul 2026

- Developed and maintained full-stack web applications using Next.js, TypeScript, Node.js, PostgreSQL, REST APIs, and Firebase Authentication, delivering production-ready features across frontend and backend systems.
- Built backend services, database integrations, authentication workflows, and AI-powered application features while collaborating with cross-functional teams on performance optimization, debugging, deployment, and product delivery.

## PROJECTS

### Plutopus — AI-Powered Predictive Network Operations Platform

#### Open Source Project [github](#) [live](#)

- Built an AI-powered Network Operations Intelligence Platform that ingests telemetry, correlates network incidents, forecasts failures, and provides topology-aware root cause analysis for proactive infrastructure monitoring.
- Engineered a distributed architecture using FastAPI, Next.js, TimescaleDB, Redpanda, Qdrant, Ollama, NetworkX, Docker, Kubernetes, Prometheus, and Grafana, implementing streaming analytics, anomaly detection, RAG-based AI Copilot, RBAC, and air-gapped deployment.

### Vayu — Autonomous Cyber Reasoning System

#### Open Source Project [github](#) [live](#)

- Developed an Autonomous Cyber Reasoning System (CRS) that discovers vulnerabilities, performs root-cause analysis, generates secure patches, and validates remediation through an automated workflow encompassing finding, reasoning, patching, and verifying to ensure robust system security.
- Built with Python, FastAPI, Ollama, Semgrep, Docker, PostgreSQL, and automated verification pipelines, incorporating static analysis, AI-driven patch generation, regression testing, provenance tracking, and fail-closed security controls.

### Weth — Zero-Trust AI Wallet using MCP

#### Open Source Project [github](#) [live](#)

- Built a Zero-Trust AI Wallet that enables AI agents to inspect balances, simulate transactions, and generate execution-ready drafts while preventing direct access to private keys.
- Implemented secure agent-wallet interactions using Model Context Protocol (MCP), TypeScript, Next.js, PostgreSQL, Ethereum, and cryptographic signing workflows, featuring transaction simulation, policy validation, audit logging, and human-in-the-loop authorization.

## SKILLS

**Programming Languages:** Java, Python, TypeScript, JavaScript

**AI Systems:** RAG, AI Agents, MCP, LLM Orchestration

**Full Stack:** Next.js, React, FastAPI, Node.js, REST APIs

**GenAI:** OpenAI, Groq, Ollama, Prompt Engineering

**DevOps:** Docker, Kubernetes, GitHub Actions, CI/CD

**Storage:** PostgreSQL, MongoDB, Redis, pgvector, Qdrant

## ACHIEVEMENTS

- HACKHAZARDS '26** (Online) – Global Buildathon • 31K+ Builders • 51 Countries **Top 25 Worldwide & Neo4j Track Honorary Mention**
- HACKSTORM 2.26** (Kolkata) – 500+ Builders **Overall Champion & Civic AI Track 3rd Place (Double Winner)**
- FrostHacks S02** (Kolkata) – 2,000+ Builders **Safety Track Winner**

## EDUCATION

**Bachelor of Technology in Computer Science Engineering (Internet of Things, Cyber Security and Blockchain Technology)**

Heritage Institute of Technology (MAKAUT)

CGPA: 8.9

**2024-2028**  
Kolkata, WB

**ISC**

St. Xavier's Institution

(CISCE)

Aggregate: 89.75%

**2022-2024**  
Kolkata, WB

**ICSE**

St. Xavier's Institution

(CISCE)

Aggregate: 92.3%

**2020-2022**  
Kolkata, WB